

Lecture 34 — Containers, Namespaces, and cgroups

Jeff Zarnett
jzarnett@uwaterloo.ca

Department of Electrical and Computer Engineering
University of Waterloo

May 30, 2026

This lecture is not yet written. The outline lives in the plan file.

Intended sections:

- 1 What containers are not
- 2 The seven namespaces
- 3 User namespaces and the UID-mapping trick
- 4 cgroups v2
- 5 Container images and OverlayFS
- 6 Runtimes vs. engines vs. orchestrators
- 7 Building a container by hand (live demo)
- 8 Escape hatches and weaknesses
- 9 When to use containers vs. VMs